EXPERIMENT 7: PRODUCTION RULES AND FORWARD CHAINING

Aim:

To implement IF-THEN production rules and infer new facts through forward chaining.

Software Required: Python 3.x

Procedure / Algorithm:

1. Initialize a set of facts.
2. Define rules as condition sets and conclusions.
3. Repeatedly apply rules whose conditions are satisfied.
4. Stop when no new facts are inferred.

Program:

```
facts = {"has_fever", "has_cough"}


rules = [
   ({"has_fever", "has_cough"}, "possible_flu"),
   ({"possible_flu"}, "recommend_medical_review")
]


changed = True


while changed:
   changed = False


   for conditions, conclusion in rules:
      if conditions.issubset(facts) and conclusion not in facts:
         facts.add(conclusion)
         print("Inferred:", conclusion)
         changed = True
```

print("\nFinal Facts:")

for fact in sorted(facts):

print(fact)Sample Input:

Initial facts:

has_fever

has_cough

Expected Output:

```
= RESTART: C:/Users/saimr/AppData/Local/Programs/Python/Python312/KE-7.py
Inferred: possible_flu
Inferred: recommend_medical_review

Final Facts:
has_cough
has_fever
possible_flu
recommend_medical_review
```

Result:

Thus, forward chaining derived additional facts by applying production rules to the initial facts.

Note: The medical-themed rules are illustrative only.